

Project#3

A Real Application Parallel Challenge

Woonki Kim

kimwoon@oregonstate.edu

8 May 2024

1. Project Environment.

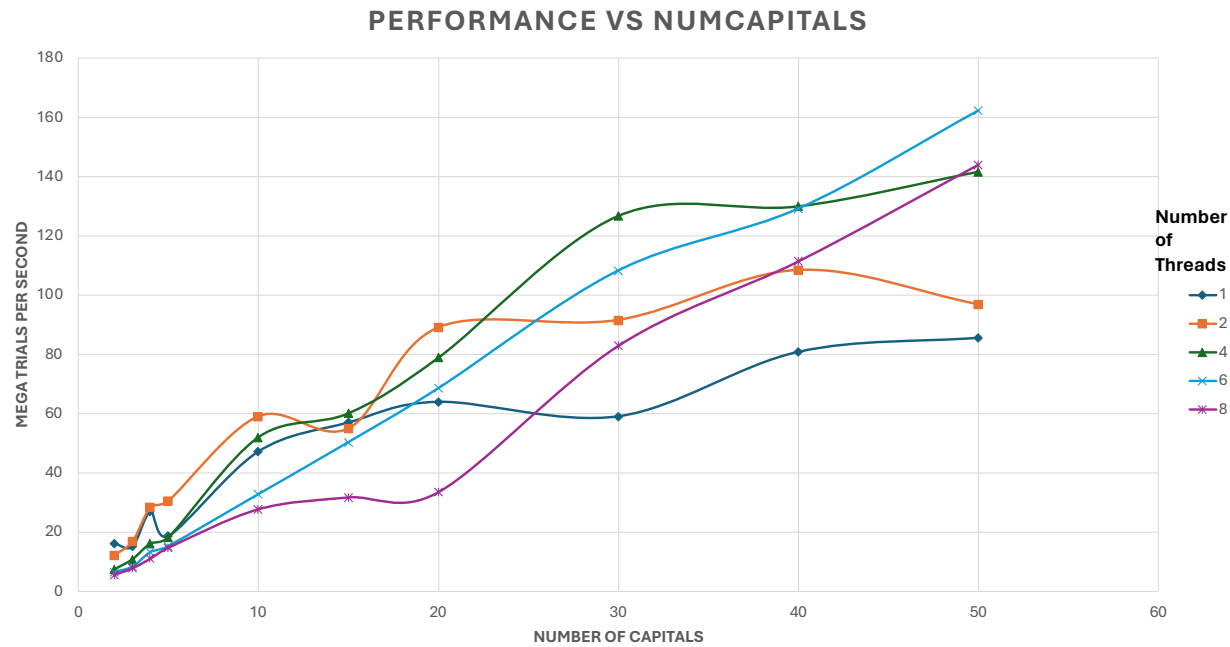
- Used FLIP2 server.
- Ran on Linux OS.
- Used g++ compiler.

2. Performance Data Table

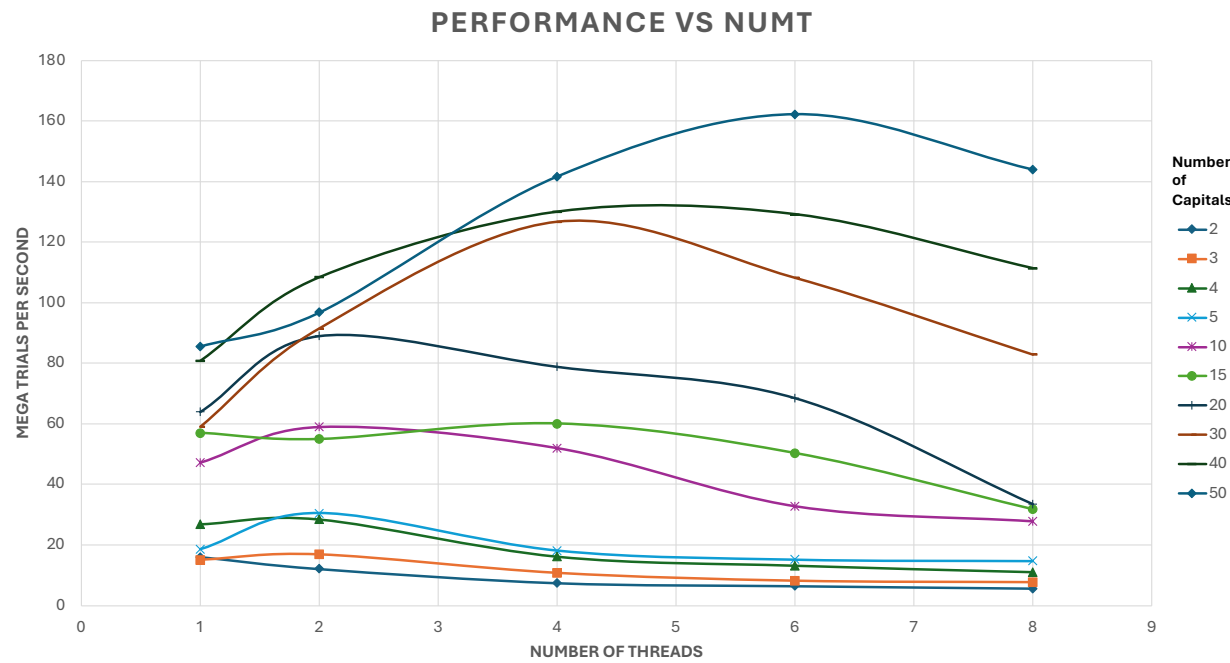
NUMCAPITALS/NUMT	1	2	4	6	8
2	16.065	12.096	7.416	6.426	5.629
3	15.109	16.858	10.844	8.294	7.822
4	26.879	28.46	16.201	13.224	11.1
5	18.671	30.527	18.217	15.23	14.756
10	47.215	59.029	52.034	32.76	27.743
15	56.998	54.933	60.086	50.306	31.72
20	63.954	89.036	78.883	68.536	33.505
30	59.078	91.524	126.765	108.26	82.904
40	80.85	108.371	129.97	129.15	111.381
50	85.6	96.768	141.654	162.231	143.969

3. Graphs

- Graph of performance vs. number of capital cities with the colored curves being NUMT



- Graph of performance vs. number of threads with the colored curves being NUMCAPITAL



4. Analysis of the Data & Graphs

- **Sweet spot at 6 threads:**
 - The performance seems to peak with 6 threads. This suggests that with 6 threads, the workload is effectively distributed, resulting in the best performance.
- **Thread 6 shows the most similar move of linear graph compared to others:**
 - Thread 6 might have a workload that is most evenly distributed, resulting in a more consistent and linear graph.
- **Increase in performance is shown when the capital number is bigger than 30:**
 - As the number of capitals increases, the workload increases. With more work to do, the benefit of parallelization becomes more apparent, resulting in increased performance.
- **Using 8 threads does not show an effective graph, it is similar to using 4 threads:**
 - The performance improvement from 4 to 8 threads is not significant.
 - Possible reasons for this could be:
 - **Diminishing returns:** The overhead of managing eight threads may outweigh the benefits of parallelism.
 - **Limited parallelism:** The workload may not be large enough to effectively utilize eight threads.

5. Extra credit.

- **Result of 50 Capitals**

CapitalNum	Longitude	Latitude	CityName	State
0	81.99	26.65	CapeCoral	FL
1	88.33	42.04	Elgin	IL
2	121.97	38.36	Vacaville	CA
3	119.04	35.35	Bakersfield	CA
4	77.62	43.17	Rochester	NY
5	105.06	39.88	Westminster	CO
6	96.77	32.79	Dallas	TX
7	79.97	32.83	Charleston	SC
8	82.35	29.68	Gainesville	FL
9	118.34	33.96	Inglewood	CA
10	118.17	33.78	LongBeach	CA
11	84.37	33.93	SandySpring	GA
12	80.28	26.08	Davie	FL
13	78.82	35.78	Cary	NC
14	122.19	47.48	Renton	WA
15	115.09	36.28	NorthLasVegas	NV
16	111.74	33.31	Gilbert	AZ
17	112.64	33.43	Buckeye	AZ
18	71.12	42.38	Cambridge	MA
19	86.79	36.17	Nashville	TN
20	98.12	29.7	NewBraunfels	TX
21	112.19	33.53	Glendale	AZ
22	116.56	43.58	Nampa	ID
23	97.35	35.24	Norman	OK
24	80.24	25.95	MiamiGardens	FL
25	119.68	36.83	Clovis	CA
26	119.25	34.27	Ventura	CA
27	117.56	34.12	RanchoCucamonga	CA
28	119.85	39.55	Reno	NV
29	121.32	38.77	Roseville	CA
30	70.94	41.66	NewBedford	MA
31	117.02	32.63	ChulaVista	CA
32	117.76	33.86	Anaheim	CA
33	122.44	45.5	Gresham	OR
34	82.07	33.37	Augusta	GA
35	80.83	35.21	Charlotte	NC
36	84.2	39.78	Dayton	OH
37	121.31	37.98	Stockton	CA
38	71.42	41.82	Providence	RI
39	76.61	39.3	Baltimore	MD
40	122.1	37.63	Hayward	CA
41	117.19	33.57	Murrieta	CA
42	94.74	39.12	KansasCity	KS
43	101.89	33.57	Lubbock	TX
44	117.14	32.81	SanDiego	CA
45	80.39	27.28	PortSt.Lucie	FL
46	106.79	32.33	LasCruces	NM
47	95.32	29.56	Pearland	TX
48	74.06	40.71	JerseyCity	NJ
49	118.18	34.69	Lancaster	CA

- **Outer loop (k):**
 - Iterates over each new capital (**NUMCAPITALS** times).
- **Inner loop (i):**
 - Iterates over each city (**NUMCITIES** times).
 - Calculates the distance (**dist**) between the current city **i** and the capital **k** using the **Distance()** function.
- **Condition check:**
 - Checks if the current city **i** belongs to the capital **k** (**Cities[i].capitalnumber == k**).
 - Checks if the distance (**dist**) between the current city **i** and the capital **k** is less than the minimum distance (**mindiscap**).
- **If condition is true:**
 - Updates the minimum distance (**mindiscap**) to the distance (**dist**) between the current city **i** and the capital **k**.
 - Updates the capital's name, longitude, and latitude to the current city's values, as this city is closer to the capital.
- **After the inner loop:**
 - The capital's name, longitude, and latitude are updated with the closest city's values.
- **Repeat the process for each capital (k).**
- In summary, this code ensures that for each capital, it finds the closest city and updates the capital's information (name, longitude, and latitude) accordingly.